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Global inequities in pesticide legislation: nearly half of pesticides approved in major crops in Latin America are not allowed in the European Union

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Pesticide use is a core strategy to control agricultural pests. Although international treaties and health recommendations call for banning hazardous pesticides, globally harmonized pesticide governance remains elusive. In Latin America (LATAM), the main net food exporter worldwide, agricultural pesticide use has increased by approximately 500% since 1990, compared with only approximately 3% in Europe.

To assess the environmental and health rigour of LATAM's pesticide legislation, we reviewed active ingredients (AIs) approved for ten major crops, and AIs banned at national level, in eight LATAM countries, assessed their hazardousness according to international standards, and legal status in the European Union (EU) and explored sources of variation in the number of AIs approved in LATAM. We identified 523 AIs approved and 236 banned in LATAM; approximately 50% of the approved and approximately 85% of the banned AIs in LATAM were either not approved or prohibited in the EU. Higher crop production and, to a lesser extent, export value were linked to more AIs approved in LATAM. These findings imply weaker regulatory frameworks for hazardous pesticides in LATAM, reflecting more permissive environmental policies compared with the EU. This reinforces the urgency of addressing asymmetries in pesticide governance and rethinking the prevailing agricultural paradigm.

1. Introduction

While essential to sustain human life, agriculture is one of the key drivers of global environmental change [1,2]. In the last seven decades, strategies to increase crop yield have involved cropland expansion [3] and conventional intensification of agriculture [4] (see Definition in electronic supplementary material), which includes monoculture production that heavily relies on external inputs [5]. Monoculture expansion and the ensuing landscape simplification have favoured the growth of pest populations [6,7], while the disruption of communities of natural enemies that feed, prey or parasitize on these pests biologically controlling their populations, has exacerbated the use of mostly synthetic pesticides for crops, also referred to as plant protection products (PPPs) [8]. In the short to the long term, heavy use of pesticides can result in negative feedbacks on crop yield because of undesired adverse effects on beneficial non-target organisms, such as insect pollinators and natural pest predators [9,10], which provide important ecosystem services to crops [11]. Furthermore, pesticides generate resistance in target organisms through strong selective pressures [12], leading in turn to increases in volume, concentration and diversification of active ingredients (AIs) used in pesticide formulations [13,14]. The outcome is a pesticide treadmill that aggravates farmers' chemical dependence [15–17].

Agricultural development shows pronounced differences among regions. Globally, the cropland area has increased by roughly 17.5% since 1961 [18]. However, during recent decades, the largest area increase occurred in the Global South (i.e. Africa, South America, Southeast Asia), while the Global North (i.e. North America, Europe) showed a sharp decrease [19]. Although dichotomous classifications (i.e. developed–developing countries or Global North–Global South) simplify the analysis of complex historical processes, perpetuating colonialist perceptions of inherent hierarchy between countries and regions [20], we resort to these classifications to preserve references to previous studies and historical contextualization. The first and second 'Green Revolutions', specifically promoted in developing countries of the Global South to meet regional and global food demands, accelerated the use of external inputs [21,22]. These regional and global pressures have probably driven a rampant increase in the volume of pesticide application [23]. In this scenario, the nearly 50% global increase in pesticide use has implied an increase of only approximately 3% in Europe but of approximately 500% in Latin America (LATAM) between 1990 and 2019 [24,25], revealing a strong contrast among regions. However, as concerning as the volume of pesticides applied in LATAM is the hazard posed by AIs used in pesticide formulations, which remains largely unexplored at the regional scale. In this study, we aim to address this knowledge gap.

Given the scientific evidence about the detrimental effects of pesticides on living organisms, human health and the environment, some AIs are classified according to hazardousness criteria, which in turn lead to restrictions or even banning of their use [25] (see Definitions in electronic supplementary material). General criteria for defining and classifying highly hazardous pesticides (HHP) include AIs contemplated in the Montreal, Rotterdam and Stockholm treaties as well as those classified in the most dangerous categories by the World Health Organization (WHO), among others (see electronic supplementary material). Although the International Code of Conduct on Pesticide Management of the Food and Agriculture Organization (FAO) remarks on the importance of harmonized pesticide legislation [25], these recommendations are unevenly transferred to the regulatory frameworks of different countries. Many countries have ratified international regulatory instruments invoked in international treaties but have not yet implemented the changes in their legislation [26]. As a consequence, many substances classified as hazardous and banned in some countries are not prohibited or are even explicitly approved in others. More specifically, developed countries have restricted or banned the use of a larger number of hazardous AIs than developing countries [26–28]. In general, AIs' approval and use across regions rely on government policies, which may be influenced by economic factors and the pressure of lobby groups [15].

The European Union (EU) is recognized for having one of the strictest regulatory systems [29], harmonized among their state members, to ensure a high level of protection for human and animal health, besides promoting biodiversity, preserving ecosystem services and securing food safety [24,30,31]. The application of the precautionary principle has progressively limited the use of verified or presumed harmful AIs (i.e. classified as not approved, registered with restrictions or banned). Notably, the recently adopted Vision for Agriculture and Food stipulates that further pesticide bans will only be enacted once viable alternatives are available [32], notwithstanding the call by European citizens for an 80% reduction in synthetic pesticide use by 2030 and a full phase-out by 2035 [33]. However, those AIs produced in Europe and elsewhere in the Global North but banned or not approved for use in the regions of production are exported elsewhere, particularly to countries in the Global South [23,34]. This reflects limited governmental and company concerns and responsibilities on the environmental and human health consequences of this asymmetric trade. On the other end, this asymmetric trade is permitted by the usually

weaker environmental legislation, plus insufficient resources to implement and enforce it, in developing countries of the Global South [26,30,35]. Additionally, dependence on foreign investment in the agricultural sector and a tendency towards large-scale agriculture have intensified pesticide use in these countries [36]. As agricultural production and exportation are important revenue sources for many of them, relaxed pesticide regulations related to crops with the highest productivity and monetary contribution to their economies could be expected.

LATAM and the Caribbean constitute the main net food exporter worldwide [37]. Agriculture accounts for more than 5% of the gross domestic product (GDP) in 20 countries of the region, with a far more significant impact through its links with other economic sectors and the development of value-added chains [37]. Moreover, agricultural exportations are a key source of foreign exchange [37,38]. The area dedicated to standing and harvesting crops in LATAM has grown by approximately 30% since 1961, representing 8.4% of its land surface today [18]. The environmental variability allows the cultivation of at least 150 different crops reported by FAO, representing approximately 20% of the primary agricultural production worldwide [18]. For instance, in 2020 LATAM contributed 70% of the global avocado production, more than 50% of sugar, coffee and soybean, among others [18]. Regarding genetically modified organism (GMO) crops, 10 LATAM countries cultivated 79.4 million ha in 2017, representing 42% of the total area globally cultivated with GMOs [39]. Nowadays, LATAM probably has the most pollinator-dependent agriculture in the world [40]. In this region, 58% of the cultivated crops have a high dependence on entomophilous pollination, while the total economic value of production for all pollination-dependent crops is estimated at 77.82 billion US dollars [41]. LATAM currently shows the highest and fastest-growing pesticide use [42]. In addition to the high volumes of pesticides used in LATAM, there is a lack of regionally unified legislation so that countries can differ in the AIs approved to treat crop pests [43].

Despite high variability among countries within each region, agricultural countries in LATAM have been historically categorized as developing countries with low to middle incomes and jointly belong to the now so-called Global South. In contrast, EU countries have been mostly categorized as developed countries with middle to high incomes and jointly belong to the Global North [20]. Therefore, associated with the economic and trade asymmetries between these two regions and contrasting environmental regulations, we first hypothesized that the pesticide regulations in LATAM are less strict than in the EU. Hence, we predicted that a considerable fraction of AIs approved in LATAM are not allowed in the EU, while all or most AIs banned in LATAM are also banned in the EU. Crops involving larger areas are more vulnerable to pests [6,44], while those crops with higher production represent important incomes for the countries' economies. In addition, crops depending on pollinators are more vulnerable to the loss of these beneficial insects [45]. These factors are expected to collectively drive the development and implementation of stricter environmental policies. Based on this expectation, we secondly hypothesized that the production, export value and ubiquity of the focal crop, as well as its pollination dependence, influence the permissiveness of pesticide regulations in LATAM. Thus, we predicted an increase in the number of AIs approved in LATAM and the number of them not allowed in the EU, with either increasing production, export value, ubiquity and non-pollinator dependence status of the target crop. To test these hypotheses, we recorded: (i) the number, identity and WHO classification of all AIs approved for use in ten of the main regional crops in eight LATAM countries; (ii) the number, identity and hazardousness classification of all AIs banned in LATAM; (iii) the EU legal status of these AIs approved or banned in LATAM; and (iv) the productive and economic variables potentially associated with the approval of AIs in LATAM.

2. Material and methods

(a) Active ingredients approved in Latin America

Data on AIs approved for agricultural use were collected from eight LATAM countries: Argentina, Bolivia, Brazil, Chile, Colombia, Costa Rica, Mexico and Uruguay. These countries accounted for approximately 87% of LATAM primary production between 2015 and 2019, which represented approximately 82% of cultivated land in the region (electronic supplementary material, table S1) [46]. Consulted sources of pesticide regulations were governmental open sources at the country level, including agricultural ministries or their dependencies, plus non-governmental open sources from the private sector (e.g. chambers of commerce) (electronic supplementary material, table S2). We recorded the number of AIs approved until the 31st of December 2020 in the selected LATAM countries in the following 10 crops, according to FAO denomination: corn, soybean, wheat, sunflower, rice, sugar cane, coffee, apple, avocado and grapes. These crops were selected due to their importance in terms of cultivated area, relative to the total cultivated land, and (or) their economic value (electronic supplementary material, tables S3 and S4) [46]. Five of these crops (i.e. soybean, sunflower, coffee, apple and avocado) depend to some extent on entomophilous pollination to produce either the seeds or fruits we consume, while the remaining five rely on anemophilous pollination [47].

We focused our search on a subgroup of PPPs [8], specifically those applied against other organisms (e.g. insecticides, herbicides, fungicides), excluding PPPs influencing the life processes of plants (e.g. growth regulators, hormones) and those preserving harvest produce, among other excluded compounds like adjuvants, surfactants and fertilizers. When information was supplied as formulations, each AI was individually recorded. Compounds were standardized by common names, International Union of Pure and Applied Chemistry (IUPAC) nomenclature [48], and Chemical Abstracts Service Registry Number [49]. Information regarding target organisms (i.e. animals, plants or fungi), WHO hazardous classification, and legal status of each AI in the EU were extracted from the EU Pesticides Database [50], the Pesticide Properties Database (PPDB) [51] and the Biopesticides Database (BPDB) [52], the last two developed by the Agriculture and Environment Research Unit at the University of Hertfordshire, UK [53]. We selected these databases because they are regularly updated and subjected to data

quality controls. The complete dataset of AIs approved for these ten crops across the eight LATAM countries is available online [46].

(b) Active ingredients banned in Latin America

We looked for regulations regarding AIs banned in the eight LATAM countries to evaluate whether hazardous AIs included in international treaties and WHO recommendations (electronic supplementary material, table S6) were banned. Information for LATAM countries was obtained from official regulations of agricultural ministries and/or sanitary authorities of each country (electronic supplementary material, table S5) [46]. For each AI banned in each LATAM country, we recorded whether they were included in the Montreal Protocol, Rotterdam and Stockholm Conventions and WHO (see electronic supplementary material, table S6 for a brief description of each international source), as well as their legal status in the EU as of 31 December 2020 [46]. We obtained the legal status in the EU of each AI from two sources: PPDB [51] and the European Chemicals Agency (ECHA) [54]. PPDB classifies each AI in one of the following categories: approved, not approved, pending and not yet assessed, while ECHA lists banned AIs. Therefore, according to the respective sources, we considered AIs as ‘not allowed’ in the EU if they were either not approved or banned.

(c) Statistical analysis

Given that not all crops were cultivated in all countries (electronic supplementary materials, table S3), a Spearman correlation was conducted to evaluate a potential association between the number of crops cultivated in each country, out of the 10 evaluated, and the number of AIs approved in each LATAM country. In addition, a Pearson correlation was carried out to assess the association between the number of AIs approved in each LATAM country for each evaluated crop and the number of them not allowed in the EU. We evaluated the influence of productive, economic, ubiquity and pollination factors on the number of approved AIs in LATAM, and on the number and proportion of them not allowed in the EU. For each country, we recorded total crop production (tonnes) and export value (US dollars) of both unprocessed and processed crop products (e.g. soybean oil), as they reflect crops’ total impact on the economy. This information was retrieved from the Food and Agriculture Organization (FAO) open database [18]. We also considered two dummy variables. The first variable indicates whether each crop is cultivated in the EU or not [55] as an indicator of crop ubiquity. The second indicates whether a crop’s yield is to some extent dependent on entomophilous pollination following Klein *et al.* [47]. Pollinator dependence was considered as an indicator of environmental awareness on the value of pollination ecosystem services regarding the use of AIs. The complete dataset is available in de Groot *et al.* [46].

Data were processed and analysed using the free software R (version 4.3.0) [56]. We applied generalized linear mixed models (GLMM) using the package ‘lme4’ [57]. The total number of AIs approved in each LATAM country for each crop, as well as the number of them not allowed in the EU, was modelled with the ‘glmer.nb’ function using a negative binomial error structure to account for overdispersion ($s.d. > \text{mean}$), while the proportion of AIs not allowed in the EU was modelled with the ‘glmer’ function considering a binomial error structure. Predictive variables included as fixed factors were the continuous ‘production’ and ‘export value’, and the binaries ‘cultivation in the EU’ and ‘entomophilous pollination’. The categorical variables ‘country’ and ‘crop’ were treated as cross-random factors in all models, accounting for geographical and agronomic variations. Pure random and mixed models were evaluated, and the best fit model was selected by the lowest Akaike information criterion (AIC) using the two-AIC-unit criterion to discriminate between models, preserving only significant predictive variables ($p < 0.05$) in final models [58]. More details on statistical analyses are provided in the electronic supplementary material and the online dataset [46].

3. Results

(a) Active ingredients approved in Latin America

A total of 523 different AIs used as PPPs were approved for any of the 10 selected crops in at least one of the surveyed LATAM countries. Of these, 455 AIs were consistently approved (i.e. approved in at least one LATAM country and not banned in any other LATAM country), while 66 AIs were approved in at least one LATAM country but banned in at least one other country of the region [46] (see also *Active ingredients banned in LATAM*). The country with the highest number of AIs approved across crops was Mexico (280), followed by Chile (260), Brazil (257), Argentina (235), Uruguay (233), Costa Rica (225), Bolivia (190) and Colombia (110) (figure 1). The number of crops cultivated per country ranged from 5 to 10 (see electronic supplementary material, table S3). There was no evidence of an association between the number of crops per country and the number of AIs approved (Spearman $\rho = 0.15$, $p = 0.73$). Among the 523 AIs approved, 202 were herbicides, 155 were targeted on animals (e.g. insects, rodents), 147 were fungicides, while 18 had multi-target action.

Regarding the WHO classification, 244 out of the 523 AIs approved in at least one LATAM country (47%) were considered dangerous to some degree (i.e. from extremely dangerous to slightly dangerous; categories Ia, Ib, II and III) (figure 2a). Six approved AIs were classified as obsolete, meaning their production and use are no longer recommended due to environmental safety or toxicity concerns. This included benzalkonium chloride, chloroneb, ethaboxam, fenoxaprop ethyl, polyoxorim and benthiazole, all of them not approved or explicitly banned in the EU. In comparison, 106 approved AIs were unlikely to present an acute hazard. Finally, 167 approved AIs were not assessed by WHO, which does not necessarily mean they are safe (e.g.

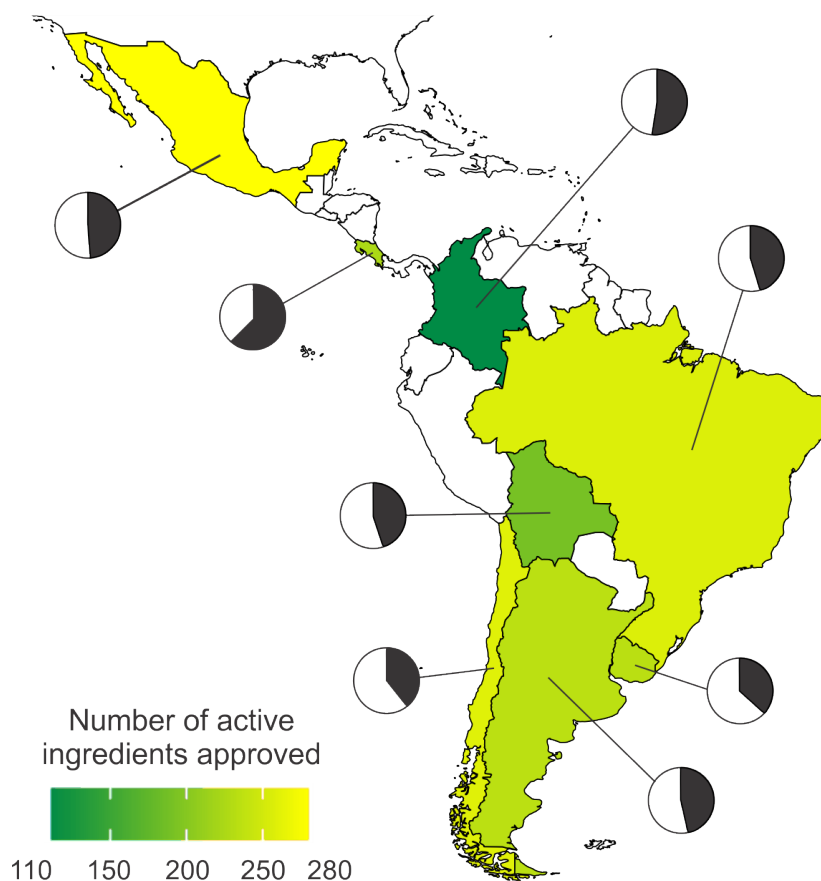


Figure 1. Total number of AIs approved for ten selected crops in eight LATAM countries until 2020. Pie charts show the proportion of them in each LATAM country that were approved (white) and not allowed (black) in the EU.

fumigants). Approximately 88% (36 of 41) of the AIs approved in at least one LATAM country that were classified in the two WHO categories of highest hazard (Ia and Ib) were not allowed in the EU (figure 2b). These included 21 organophosphates (e.g. methidathion, parathion-methyl), seven anticoagulant rodenticides (e.g. coumatetralyl, diphacinone), four carbamates (e.g. aldicarb, carbofuran), three pyrethroids (e.g. beta-cyfluthrin, cyfluthrin) and one organochlorine (edifenphos). In turn, 50% (157 out of 311) of the AIs approved in at least one LATAM country belonging to WHO categories of less hazardousness (II, III, U) were not allowed in the EU (figure 2b). The AIs in category II encompassed 24 chemical families. The most represented among these were organophosphates (15 AIs; e.g. chlorpyrifos, ammonium glufosinate, acephate), carbamates (10 AIs; e.g. carbaryl, carbosulfan, cartap, thiram), pyrethroids (seven AIs; e.g. cyhalothrin, fluvalinate, permethrin), triazoles (seven AIs; e.g. cyproconazole, propiconazole), quaternary ammonium compounds (four AIs; e.g. diquat, paraquat), neonicotinoids (three AIs; e.g. imidacloprid, thiacloprid) and organochlorines (three AIs; e.g. endosulfan, fomesafen; for more details, see [46]).

When comparing the total number of AIs approved in LATAM with EU legislation, we found that 256 of 523 AIs approved in at least one LATAM country (48.9%) were not allowed in the EU. Among the 455 AIs consistently approved in all the eight countries of LATAM surveyed, 199 (43.7%) were not allowed in the EU. Costa Rica had the highest number of approved AIs that were not allowed in the EU (140), followed by Mexico (135), Brazil (115), Argentina (106), Chile (99), Bolivia (87), Uruguay (86) and Colombia (57) (figure 1). Relative to the total number of AIs approved in each LATAM country, Costa Rica presented 62% of AIs not allowed in the EU, followed by Colombia with 52%, while the rest of the countries were slightly below 50% (figure 1). On average, 46.6% of the AIs approved in a LATAM country were not allowed in the EU. The number of AIs approved in LATAM per crop and country presented a marked positive association with the number of them not allowed in the EU (Pearson's $r = 0.95$; $n = 65$; $p < 0.001$; electronic supplementary material, figure S1). The number of AIs approved in LATAM by country and crop and the number of those AIs not allowed in the EU are depicted in electronic supplementary material, figure S2.

(b) Active ingredients banned in Latin America

Regardless of crop, a total of 236 AIs were banned in at least one LATAM country (range among countries = 13–98; figure 3). Of these, 157 AIs were consistently banned across countries (i.e. banned in at least one LATAM country and not approved in any other LATAM country), while 66 AIs were banned in at least one country but approved in at least one other LATAM country. The highest number of AIs banned was found in Costa Rica (98), Argentina (93) and Brazil (86), while the remaining countries ranged between 13 and 30 AIs banned. In total, 202 out of the 236 AIs (85%) banned in LATAM countries were also either not approved (142) or banned (60) in the EU (figure 3).

Latin American countries varied in the number of AIs banned that were included in international treaties and WHO recommendations (figure 4) [46]. AIs included in the Montreal Protocol (total number of AIs = 2) were only banned in Costa

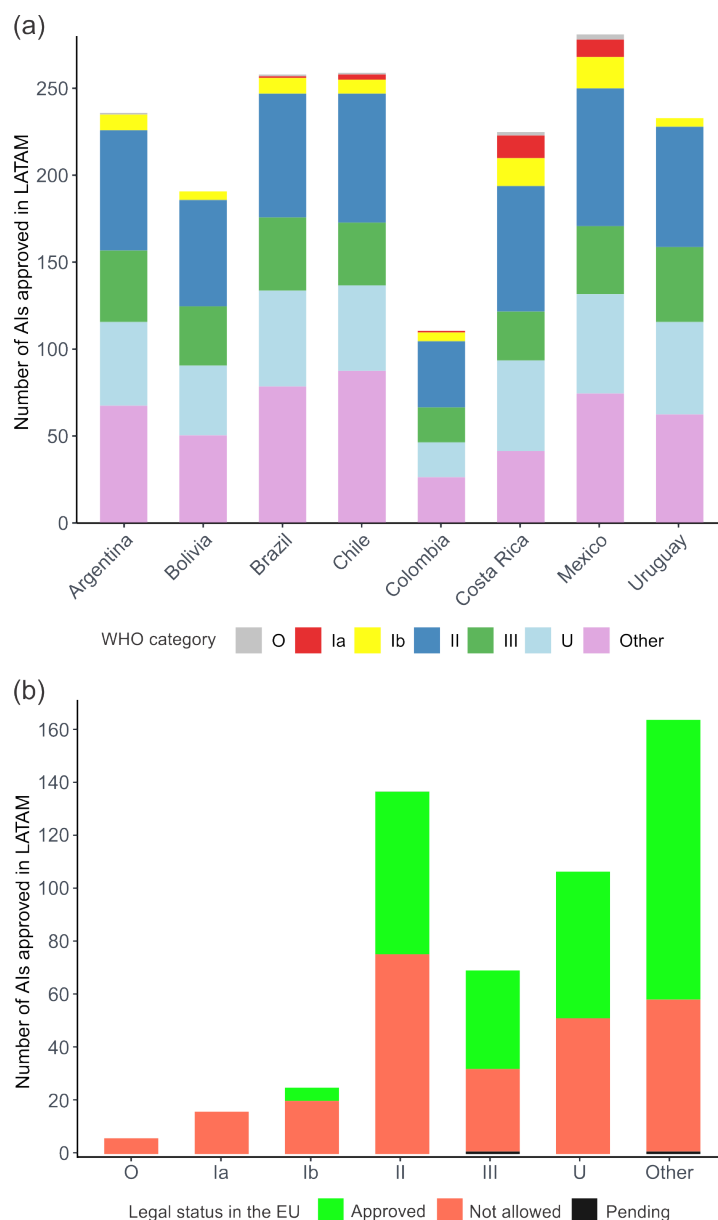


Figure 2. AIs approved in LATAM until 2020, classified following the World Health Organization (WHO) recommendations, (a) grouped by country and (b) legal status in the EU. WHO category Ia = extremely hazardous, Ib = highly hazardous, II = moderately hazardous, III = slightly hazardous, U = unlikely to present acute hazard, O = obsolete. The category 'Other' includes AIs that were not assessed by WHO.

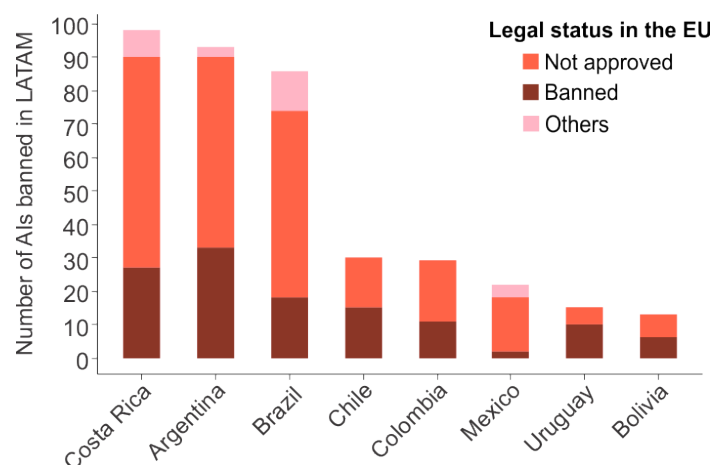


Figure 3. Total number of AIs banned in each LATAM country until 2020 and their legal status in the EU, according to the Pesticides Properties DataBase (PPDB) and the European Chemicals Agency (ECHA) reports. The category 'Others' includes AIs that were either approved or not listed in the EU.

Rica (1) and Colombia (1). In the case of AIs included in the Rotterdam Convention (total AIs = 45), the countries that had more AIs banned were Argentina (28) and Costa Rica (27), followed by Chile (22). For those AIs included in the Stockholm

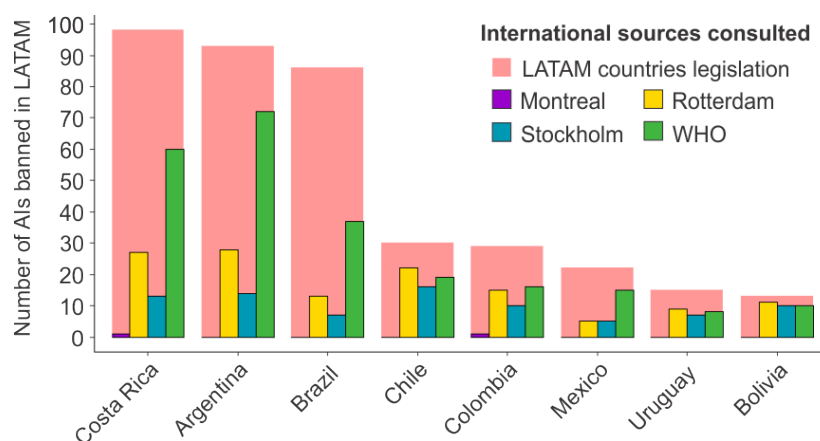


Figure 4. Total number of AIs banned in each LATAM country until 2020 and number of them that are included in international treaties and/or health recommendations because of their hazardousness to the environment and/or human health. For WHO, only categories of the highest hazard (Ia, Ib and O) were considered.

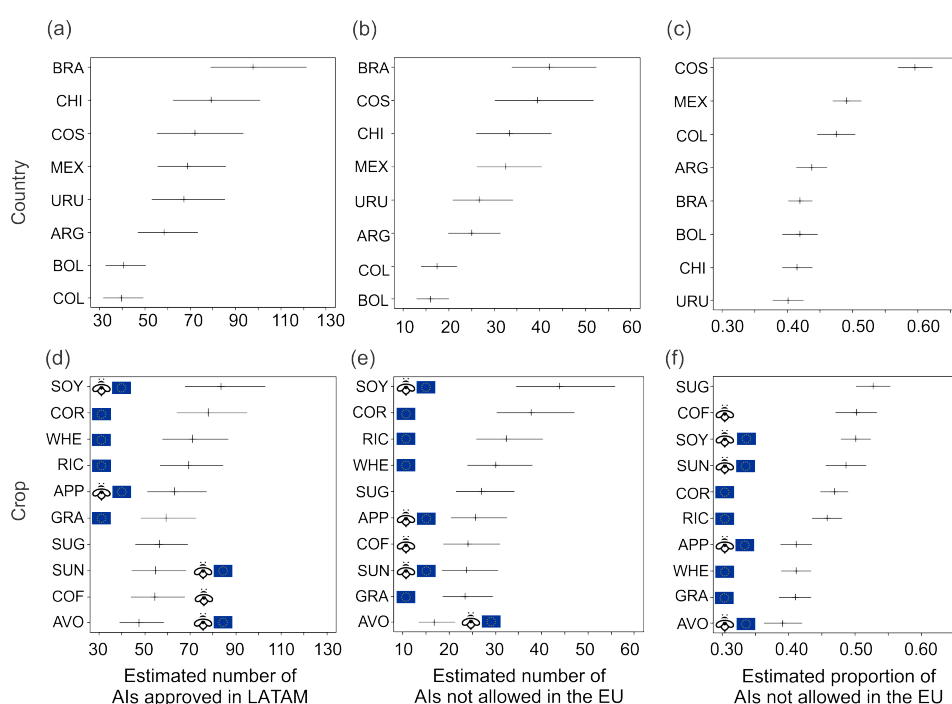


Figure 5. Estimated values based on pure random models of the number of AIs approved in LATAM until 2020, and number and proportion of them not allowed in the EU, by country (a, b and c) and crop (d, e and f). The horizontal lines represent the standard error of each estimate. Crops that rely on entomophilous pollination are denoted by an insect symbol, while crops that are cultivated in the EU are denoted by the EU blue flag. ARG = Argentina; BOL = Bolivia; BRA = Brazil; CHI = Chile; COL = Colombia; COS = Costa Rica; MEX = Mexico; URU = Uruguay. APP = apple; AVO = avocado; COF = coffee; COR = corn; GRA = grape; RIC = rice; SOY = soybean; SUN = sunflower; WHE = wheat.

Convention (total AIs = 18), again, Chile (16), Argentina (14) and Costa Rica (13) led the list. Regarding AIs classified by WHO in the most dangerous categories (Ia and Ib) plus those classified as obsolete (O) (total AIs = 123), Argentina (72) and Costa Rica (60) had more AIs banned, while the other LATAM countries had between 8 and 37.

(c) Active ingredients approved in Latin America and not allowed in the European Union: sources of variation and predictors

The pure random model showed that 32% of the variability in the number of AIs approved in LATAM was attributable to the two random factors: 20.3% explained by variation among countries and 11.8% by variation among crops (electronic supplementary material, table S7). Brazil and Chile had the highest estimated number of approved AIs, while Bolivia and Colombia presented the lowest number (figure 5a). Regarding crops, soybean and corn exhibited the highest estimated number of approved AIs in LATAM countries, and avocado the lowest (figure 5d). When fixed factors were included, the model that best explained the variance of the data included crop production ($\beta_1 = 0.33$; 95% CI = 0.10–0.56; $p = 0.006$), exportation value ($\beta_2 = 0.23$; CI = 0.02–0.43; $p = 0.029$) and its cultivation in the EU ($\beta_3 = 0.69$; CI = 0.28–1.10; $p = 0.001$). Based on these estimates, an increase of one order of magnitude in production was estimated to lead to the approval of 20.7% additional AIs in LATAM.

Similarly, an equal increase in export value resulted in the approval of 2.4% additional AIs. For crops also cultivated in the EU, the model estimates indicated a 99% increase in AIs approved in LATAM compared with crops not cultivated in the EU. These three fixed factors accounted for all the variability observed among crops (electronic supplementary material, table S7). We found no evidence that crop dependence on entomophilous pollination explains any variability in the number of AIs approved in LATAM.

Regarding the number of AIs approved in LATAM but not allowed in the EU, the pure random model indicated that 35% of the variability was due to random factors: 21% due to variation among countries and the remaining 14% due to variation among crops. In this case, Brazil and Costa Rica had the highest estimated number of AIs approved that were not allowed in the EU, while Colombia and Bolivia presented the lowest (figure 5b). Regarding crops, again, soybean and corn had the highest estimated number of AIs approved that were not allowed in the EU, while avocado had the lowest (figure 5e). The mixed model that best explained the variability of the data included the fixed effects of crop production ($\beta_1 = 0.28$; CI = 0.06–0.51; $p = 0.015$), crop exportation value ($\beta_2 = 0.30$; CI = 0.09–0.51; $p = 0.005$) and its cultivation in the EU ($\beta_3 = 0.47$; CI = 0.05–0.88; $p = 0.028$). We estimated that an increase of one order of magnitude in production resulted in the approval of 17.3% more AIs in LATAM that were not allowed in the EU, while a similar increase in the exportation resulted in the approval of 3.1% additional AIs in LATAM that were not allowed in the EU. For crops also cultivated in the EU, estimates indicate an increase of 60% in the number of AIs approved in LATAM that were not allowed in the EU, compared with crops not cultivated in the EU. The fixed factors accounted for all the variability observed among crops (electronic supplementary material, table S7). As before, we found no evidence that crop dependence on entomophilous pollination explains variability in the number of AIs approved in LATAM that were not allowed in the EU.

The random model for the proportion of AIs approved in LATAM and not allowed in the EU indicated that 75% of the variability in the data was explained by random factors, with 44% due to variation among countries and the remaining 31% among crops. Costa Rica had the highest estimated proportion of AIs approved that were not allowed in the EU, followed by Mexico and Colombia, while Uruguay had the lowest (figure 5c). Concerning crops, sugar cane and coffee, the only two crops that were not produced in the EU presented the highest estimated proportions of AIs approved that were not allowed in the EU, together with soybean, while avocado presented the lowest estimated proportion (figure 5f). According to the mixed model, cultivation in the EU was the only relevant fixed factor ($\beta_3 = -0.39$, CI = $(-0.69) - 0.09$, $p = 0.011$) (electronic supplementary material, table S7). This means that for crops cultivated in the EU, the probability of AIs being not allowed in the EU is estimated at 43%, whereas for crops not cultivated in the EU, this probability increases to 53%. Finally, we found no evidence that production, export value and crop dependence on entomophilous pollination explain variability in the proportion of AIs approved in LATAM that were not allowed in the EU.

4. Discussion

The revision of approved and banned AIs for PPPs in LATAM reveals strong disparities in pesticide legislation, not only in comparison with the EU but also among countries within the region. Considering only the products approved for the ten crops and eight countries analysed, from a total of 523 AIs approved in at least one of the countries surveyed (ranging from 110 to 280), 47% were classified by WHO in categories ranging from extremely dangerous (Ia) to slightly dangerous (III), 12.6% (66) were banned in other countries of the region, and approximately 50% (256) were not allowed in the EU (figure 1). Considering the banned substances regardless of crops, from a total of 236 AIs banned in at least one LATAM country, almost 28% (66 AIs) were approved in other countries in the region, while 85% were also banned in the EU. The number of AIs approved in LATAM strongly correlated with the number of them not allowed in the EU, and while both values varied considerably among countries, and to a lesser extent among crops, most of the among-crop variation was explained by crop production and export value, and whether the crops were also cultivated in the EU. Here, we discuss potential underlying drivers for this scenario, including weak and contradictory pesticide regulations in LATAM, the influence of economic factors, and the implications for human health, environment and trade, as well as alternatives for a global governance of pesticides.

Many AIs assessed by WHO as HHP because of their high levels of acute or chronic hazards to health or environment that are also regulated by international treaties were not banned or, more concerning, were explicitly approved in LATAM countries. The fact that 28% of all banned AIs were approved in at least one country while banned in at least another country reveals contrasting criteria in pesticide regulations among countries and reflects the lack of harmonized legislation at the regional level. Furthermore, the finding that no LATAM country had banned all HHPs revised in this study indicates a gap in the implementation of international treaties and conventions within local legislation [30]. This is particularly worrisome, as both direct and indirect human exposure to pesticides disproportionately impacts developing countries of the Global South [59]. Moreover, since pesticide volumes larger than indicated are frequently applied in developing countries due to either a lack of knowledge or poor recommendations of pesticide traders [60], the risk of exposure to HHP is expected to be even higher. One possible explanation is that older and more hazardous AIs, already banned in some countries, are sold at lower prices where legislation is less rigorous [26], but data on cross-country AI market prices remain scarce.

Notable differences were found between LATAM countries and the EU in the rigour applied to regulate pesticides. Nearly half of the AIs approved in at least one LATAM country were not allowed in the EU, while most AIs banned in LATAM were also not allowed in the EU, reinforcing disparity in pesticide legislation between these two regions. As a consequence, the set of AIs approved and potentially applied in crops differed significantly between both regions. According to our review, examples of AIs approved in LATAM but not allowed in the EU were, for instance, insecticides like acephate and chlorpyrifos, both organophosphates highly toxic to humans through neurotoxicity and genotoxicity, presenting unacceptable risks to operators

and consumers, neonicotinoids and fipronil, extremely toxic to bees and other pollinators, or carbaryl, a suspected carcinogen with high acute and long-term toxicity to birds, mammals, aquatic life and beneficial arthropods. Herbicides, such as diquat and paraquat, were approved in LATAM, which can cause severe human poisoning through skin contact or inhalation, with paraquat being lethal even at very low doses, and both are highly toxic to aquatic organisms. This difference in pesticide regulation also sets the scenario where the transnational pesticide industry can push to keep its chemical products in the LATAM market once they are banned in the EU or elsewhere [23,34]. In this context, it is expectable that most of the AIs approved in LATAM and classified as most dangerous by WHO (i.e. Ia, Ib, O) were not allowed in the EU, while those less hazardous (i.e. II, III, U) were more broadly approved in the EU. Since the number of AIs approved in LATAM strongly correlated with the number of them not allowed in the EU (i.e. either not approved or banned), the number of AIs approved for a particular country and crop in LATAM provides an idea of the magnitude of hazardous AIs potentially applied.

The number of AIs approved in LATAM, as well as the number of them not allowed in the EU, varied across crops and countries in LATAM. This variability could be related to at least two main non-exclusive causes: technical/agronomic needs (i.e. differences in the environmental conditions influencing pest incidence and/or impact on crops) and economical and political factors (i.e. decision-making processes regarding regulations as a function of the economic relevance of a given crop). When crop production, export value and its cultivation in the EU were included as predictive variables in the statistical models, they explained practically all the variation among crops. This suggests that economy and trade, led by whether or not a crop is cultivated in both regions and its production volume, played a more prominent role influencing AIs' approval in LATAM regardless of agronomic considerations. However, as agronomic variables were not included in our model, we cannot conclusively assess their relative importance. Specifically, economically significant crops in terms of production and export value in LATAM, that were also cultivated in Europe implying an amplified pesticide market, had more AIs approved to use and a greater number of them not allowed in the EU. Finally, we found no evidence of environmental policies specifically oriented to protect organisms beneficial to crops like pollinators in LATAM, as our analysis did not provide evidence that the number of AIs approved differed between entomophilous and non-entomophilous pollinated crops.

According to our results, soybean and corn, two of the most cultivated GMO crops in LATAM, had the highest estimated number of AIs approved and the highest number of them not allowed in the EU, followed by wheat and rice. This finding deserves special attention, as GMOs play a central role in LATAM agriculture. During the 1990s, GMO crops were introduced in the region, and just within 20 years, the area cultivated with GMO accounted for nearly half of the global total for such crops [39]. In contrast, agricultural production with GMOs is banned in most EU countries [61]. GMOs are typically associated with technological packages that include the massive use of pesticides, such as glyphosate-based herbicides to which GMO soybean and corn are resistant. In this context, our results raise concerns, particularly in light of the recent approvals in Argentina, Brazil and Paraguay of the first GMO wheat resistant to drought and ammonium glufosinate [62], a crop that is already among the top three with the highest estimated number of AIs approved in LATAM (figure 5d) before GMO varieties were released. Our findings challenge the claim that GMOs would reduce pesticide dependency. A critical evaluation of their use is urgently needed, particularly given their prevalence in conventional intensive agricultural systems.

Despite this gloomy scenario, there are some counter-reactive trends in LATAM. The conventional intensification of agriculture, with its partial and simplified short-term view of agroecosystems and reliance on molecular and chemical solutions, seems to be giving way, at a low pace, to more holistic management approaches that embrace the economic, ecologic and social complexities of agroecosystems. Agroecology, which avoids pesticides and GMOs, offers a sustainable alternative by promoting diversified, productive and resilient agricultural landscapes [63,64]. Emerging in LATAM during the 1960s and 1970s as a response to the agro-industrial model, agroecology has grown over recent decades [63,65]. This agricultural approach aims at increasing sustainable production while conserving natural resources, making agroecology the most promising approach for managing agroecosystems [66]. However, despite recent laws and policies promoting this more environmentally friendly approach, implementation efforts and adoption still remain fragmented [67].

(a) Concluding remarks and recommendations

Our results underscore the varying approaches to the implementation of the precautionary principle and international treaties across different countries, reflecting important differences in the protection of human, animal and environmental health. In the case of LATAM, a region within the Global South, these results also reveal a critical weakness, as hundreds of AIs harmful to both human health and the environment remain approved. As shown, the approval of pesticides is shaped by productive and economic factors in countries whose economies rely on agricultural exports and have recently undergone a process of economic 'reprimarization' [68]. This makes it both urgent and challenging to strengthen environmental regulations on pesticide use. Moreover, the influence of the agricultural economic sector on different spheres of governance in LATAM [43,69], combined with the close ties between the states and the multinational capital and the ensuing lobbies, imposes significant constraints on the capacity of national governments to act independently [68].

Pesticides are a major global contamination source, increasing in volume and chemical diversity [70–73], including hazardous and banned compounds that accumulate in the environment [35,74,75]. Rural workers are continuously exposed [76], while living near pesticide application areas potentiates the incidence of different types of cancers and other lethal diseases [77]. Even household dust from both conventional and organic farms can carry hazardous pesticide residues [78], while urban populations face exposure through contaminated food and water [79,80]. Interestingly, food from areas with minimal pesticide use may also contain harmful residues [81], highlighting how widespread pesticides are. In addition to the impact on human health, massive

pesticide use and its diversity represent a menace to the wild biota remaining in agroecosystems [82,83], which can jeopardize different related ecosystem services like crop pollination [42].

A critical evaluation of the trade-offs associated with agricultural pesticide use requires a holistic approach that considers not only the economic costs but also the social and environmental impacts [23]. As an immediate priority, the production, commercialization and use of all HHP should be banned worldwide to prevent trade asymmetries [84]. Strengthening collaboration between scientists and decision-makers is essential, as the mobilization of expertise significantly influences pesticide governance [85]. Risk management at local and regional scales should be enforced in pesticide legislation, ensuring that the approval of PPPs aligns with updated risk assessment protocols [86] and monitoring schemes, which are context-dependent [87,88]. Risk assessments must also improve to account for environmentally relevant mixtures [75] and, more rigorously, the effects on non-target species [89,90]. Policies should include official, rigorous open data sources, pesticide monitoring programmes, increased access to reliable information for workers, consumers and communities, training on integrated pest management and incentives to promote safer agricultural practices such as agroecological farming. Urgent, ambitious, and collaborative efforts are required from all stakeholders and countries. We hope this contribution helps to improve pesticide legislation in LATAM.

Ethics. This work did not require ethical approval from a human subject or animal welfare committee.

Data accessibility. Agricultural data, active ingredients data and R codes: CONICET Repository [46].

Supplementary material is available online [91].

Declaration of AI use. We have not used AI-assisted technologies in creating this article.

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